

Ex. 2

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```
print(uuid::UUIDgenerate())
```

```
## [1] "dde9fdb7-2d6e-40bf-b5be-ed6ab3b6af3d"
```

```
library(ggplot2)
```

Data for Exercise 1 and 2 :

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,  
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)
```

Question 1

Step 1

```
highest <- max(commute)  
lowest <- min(commute)  
data_range <- highest - lowest  
class_width <- ceiling(data_range / 5)  
  
c(Highest = highest, Lowest = lowest,  
  Range = data_range, Width = class_width)
```

```
## Highest Lowest Range Width  
##      45      18      27      6
```

Step 2 and 3

```
library(dplyr)
```

```
class_breaks <- seq(17.5, 47.5, by = class_width)  
class_limits <- c("18-23", "24-29", "30-35", "36-41", "42-47")  
commute_freq <- data.frame(commute) |>  
  mutate(Class = cut(commute, breaks = class_breaks, labels = class_limits)) |>  
  count(Class, .drop = FALSE) |>  
  rename(Frequency = n) |>  
  mutate(  
    Boundaries = paste(head(class_breaks, -1), "-", tail(class_breaks, -1)),  
    Midpoint = head(class_breaks, -1) + class_width / 2,  
    Cumulative = cumsum(Frequency)  
  )
```

Output table

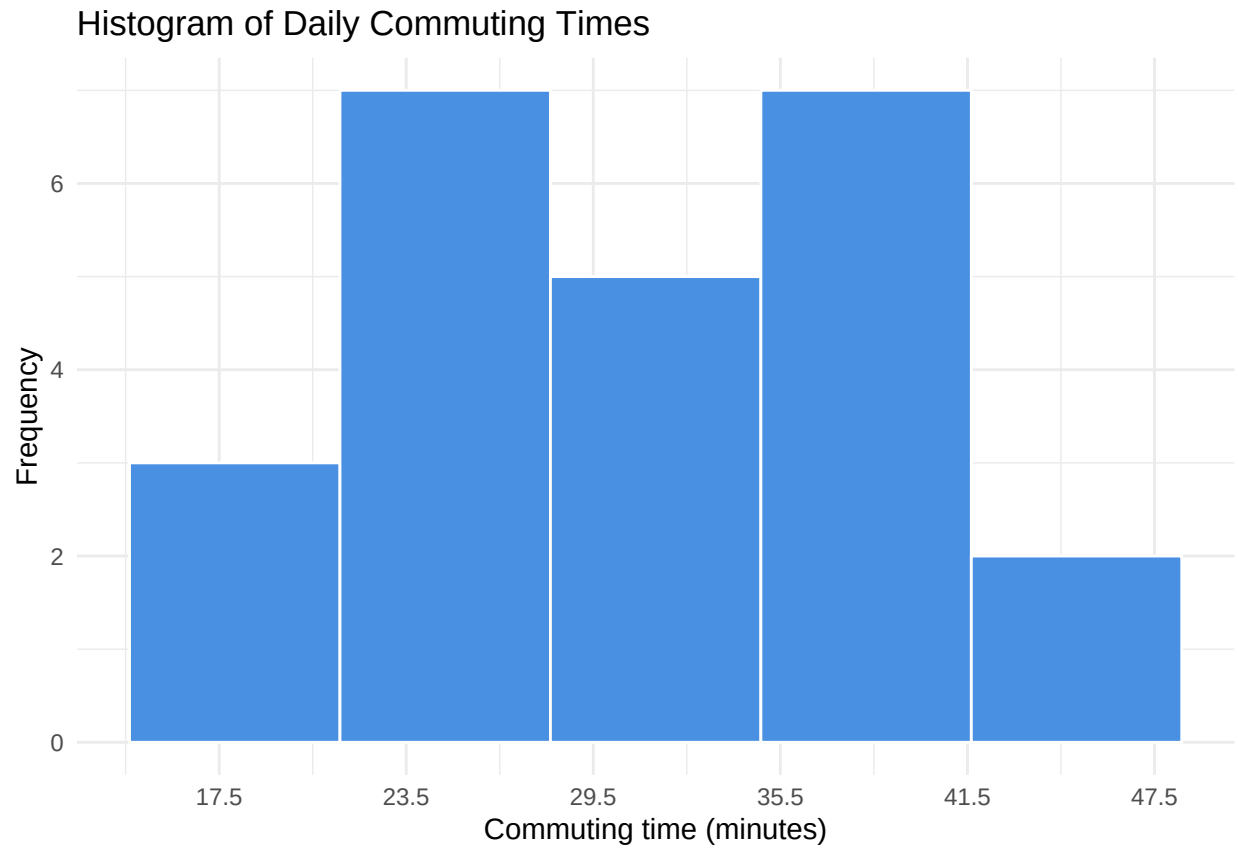
```
knitr::kable(  
  commute_freq[, c("Class", "Boundaries", "Midpoint", "Frequency", "Cumulative")],  
  col.names = c("Class limits", "Class boundaries", "Midpoint", "Frequency", "Cumulative"),  
  align = "c"  
)
```

Class limits	Class boundaries	Midpoint	Frequency	Cumulative
18-23	17.5 - 23.5	20.5	5	5
24-29	23.5 - 29.5	26.5	6	11
30-35	29.5 - 35.5	32.5	5	16
36-41	35.5 - 41.5	38.5	6	22
42-47	41.5 - 47.5	44.5	2	24

```
library(ggplot2)
```

Question 2

```
ggplot(data.frame(commute), aes(x = commute)) +  
  geom_histogram(bins = 5, fill = "#4a90e2", color = "white") +  
  scale_x_continuous(breaks = class_breaks) +  
  labs(  
    title = "Histogram of Daily Commuting Times",  
    x = "Commuting time (minutes)",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```



The distribution is approximately bell-shaped and symmetric, with a single central peak and no apparent outliers.